

Problem 1

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Problem 1 For the following problems, the derivative is given. Determine which function was the original function.

(a) The derivative is $f'(x) = \cos(x)e^{\sin(x)}$. Which is the original function?

(i) $f(x) = (\sin(x))(e^x)$

(ii) $f(x) = \sin(e^x)$

(iii) $f(x) = e^{\sin(x)}$

(iv) $f(x) = e^{x \sin(x)}$

(b) The derivative is $g'(x) = 4(\tan(x^4 - 5x))^3 \sec^2(x^4 - 5x)(4x^3 - 5)$. Which is the original function?

(i) $g(x) = (\tan(x) - 5x)^4$

(ii) $g(x) = \tan^4(x) - 5x^4$

(iii) $g(x) = \tan(x^4 - 5x)$

(iv) $g(x) = \tan^4(x^4 - 5x)$